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A novel microbubbles-assisted Fe-C micro-electrolysis system to bridge the anaerobic digestion and microalgae cultivation: the dual drive of electron transfer and coagulation

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ABSTRACT

Microalgae-based biogas slurry treatment systems, which possess the dual capability of assimilating nutrients and sequestering CO_2, are widely recognized as a promising technology. However, the complex composition of biogas slurry hinders microalgal metabolic activity and limits the system's efficiency. In this study, a novel microbubbles-assisted Fe-C micro-electrolysis (MFC) system was proposed for biogas slurry pretreatment to overcome the bottleneck of utilizing microalgae in biogas slurry treatment. The MFC system significantly improved the electron transfer rate, which facilitated the oxidation of Fe(II) to Fe(III) and achieved self-accelerating coagulation, increasing the removal efficiencies of dissolved organic matter (DOM) and turbidity to 37.3% and 94.9%, respectively. The results revealed that rapid oxidation and continuous perturbation by microbubbles caused the formation of more single-corner coordination between FeO6 octahedrons, as well as inhibited the transformation of hydrotalcite to the highly crystalline mineral. The accumulation of hydrotalcite significantly enhances the adsorption of specific types of DOM by forming a more stable structure in the flocs. In addition, the MFC system effectively removed stressors (e.g., humic-like substances, aromatic compounds, and residual iron ions) of microalgae in biogas slurry and increased the maximum quantum conversion efficiency (Fv/Fm) of the photosynthetic system from 0.455 to 0.556. Consequently, the biomass of microalgae cultured in the biogas slurry treated by the MFC system was increased by 42.7% compared to the conventional Fe-C micro-electrolysis (CFC) system. Overall, this study demonstrates that MFC-driven oxidative synergistic self-accelerating coagulation is a promising process for the stabilization and sustainability of microalgae-based biogas slurry treatment.

1. Introduction

Biogas slurry, a byproduct of anaerobic digestion, is rich in organic compounds and nutrients, which pose potential threats to the ecosystem if not managed properly [13]. In China, the discharge of biogas slurry is substantial due to the widespread adoption of anaerobic digestion for farm wastewater treatment. Among them, the swine wastewater discharge has escalated to 228.3 million with an assumed volume of 5.26 t wastewater/pig [31]. The residual biogas slurry remains a source of environmental concern due to its high content of bioavailable nutrients, including nitrogen and phosphorus. Common methods of biogas slurry treatment include artificial wetlands [14], membrane filtration [38], struvite crystallization [7], and aerobic treatment. However, these methods still face limitations such as low efficiency, high cost, poor scalability, or high energy consumption in practical applications.

The integration of microalgae cultivation with biogas slurry treatment presents a sustainable system for recycling nutrients from biogas slurry. Moreover, it offers numerous additional advantages, including

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nutrient recovery, carbon sequestration, and value-added biomass production [2,36]. However, the complex composition of biogas slurries may include animal manure, food waste, and agricultural residues, all of which can impact microalgae cultivation [13]. For instance, agricultural residues contain lignocellulose, which contributes to high turbidity and is difficult to degrade. Food waste may introduce additives such as heavy metals and antibiotics, while animal manure typically contains elevated levels of ammonia nitrogen.

Before microalgae cultivation, pretreatment is typically employed to reduce inhibitory factors in biogas slurry. For instance, centrifugation effectively removes suspended solids and colloidal particles, thereby reducing the turbidity of biogas slurry. However, its high energy consumption limits its scalability for large-scale applications [36]. Filtration offers a low-energy alternative, but filter clogging and frequent maintenance remain challenges [38]. Other methods, such as flocculation, dilution, and advanced oxidation processes, also require balancing efficiency and operational costs [1,32]. Hence, there is an urgent need for an advanced pretreatment technology with engineering potential.

Fe-C particles have been widely used in wastewater treatment owing to their simplicity, cost-effectiveness, and sustainability [11,20]. The Fe-C micro-electrolysis accompanied by oxidation, reduction, adsorption, and coagulation, shows enormous potential in the removal of harmful pollutants (e.g., heavy metals, polycyclic aromatic hydrocarbons, humic acids, etc.) from biogas slurry [15]. Nevertheless, the low mass transfer efficiency and narrow pH range of conventional Fe-C micro-electrolysis limit the electron transfer rate, making the removal of pollutants unsatisfactory [15,39]. These issues directly impede the system's ability to pretreat biogas slurry effectively for microalgae cultivation. While enhancements like the addition of chemicals [10], magnetic fields [39], and structural improvements [29]have been attempted, the operational complexity and scalability limitations make these methods impractical for large-scale applications.

In addition, the turbidity of the biogas slurry can be effectively removed by coagulation, which is essential for microalgae cultivation. However, there are few studies on the coagulation enhancement mechanism of Fe-C micro-electrolysis. Fe(III) produced by the oxidation of Fe(II) can lead to the precipitation of colloidal-sized (0.02–0.2 μm) Fe (hydr)oxides, like ferrihydrite [26]. Ferrihydrite can adsorb organic functional groups in the biogas slurry and subsequently aggregate to form sinking authigenic particulate Fe, due to its high reactivity and expansive specific surface area [24]. However, the Fe-C micro-electrolysis system tends to be acidic, thus lowering the oxidation rate of Fe (II) [39]. Furthermore, ferrihydrite is a poorly crystalline mineral and undergoes a gradual transformation into highly crystalline minerals, compromising their capacity for adsorption and coagulation performance of pollutants [30]. Hence, further investigation is needed to enhance the rate of Fe-C micro-electrolysis and improve the coagulation performance for pollutant removal of the biogas slurry.

Recently, microbubbles (with diameters < 100 μm), with high mass transfer efficiency and robust interface effect, have gained extensive application in wastewater treatment [23]. These properties have the potential to accelerate the rate of electron transfer in Fe-C micro-electrolysis processes. Microbubbles-assisted aeration has also been reported to be advantageous in reducing energy consumption and costs [8]. Additionally, previous studies have found that a substantial quantity of hydroxyl radicals (•OH) are generated during the collapse of microbubbles [9], which can accelerate both pollutant degradation and the oxidation of Fe(II) to Fe(III). In situ formation of Fe(III) species by oxidizing Fe(II) is typically more effective in water treatment compared to the direct addition of Fe(III) [33]. These studies indicate the potential contribution of microbubbles to the efficiency of conventional Fe-C micro-electrolysis treatment of biogas slurry.

To address the dual challenges of sluggish electron transfer and incomplete coagulation in conventional Fe-C systems, this study proposes a novel microbubbles-assisted Fe-C micro-electrolysis (MFC) system. The MFC was applied to biogas slurry pretreatment to make it more suitable for microalgae cultivation. Specifically, the performance of removing pollutants and different types of dissolved organic matter (DOM) by degradation and coagulation was evaluated. Additionally, the mechanism related to the microbubbles enhancing the electron transfer rate of micro-electrolysis was illustrated. Furthermore, the effect of microbubbles on the coordination structure of Fe atoms and the crucial role of various Fe minerals transformations in the coagulation process for DOM removal were analyzed. Finally, the improvement of microalgae culture performance by pretreated biogas slurry and the feasibility of nutrient utilization were investigated. Overall, the proposed process can effectively improve the stabilization and sustainability of microalgae-based biogas slurry treatment systems, demonstrating great potential for supporting the transition to sustainable water treatment.

2. Materials and methods

2.1. Raw materials and swine biogas slurry

Fe-C particles (3–6 mm in diameter) were obtained from Green Source Activated Carbon Co., Ltd. China. The characterization details are given in Text S1 and Fig. S1. Microbubbles with a mean size of 31.16 $\pm$ 1.32 μm and a concentration of 1.63×10^6 bubbles/mL (Fig. S2), were generated by a microbubble generator (ZJC-NM-200L, Zhongjing, China). Conventional bubbles with an average size of about 1 mm, were produced using a commercially available cylindrical bubble stone aeration.

Swine biogas slurry was collected from the anaerobic digester effluent of a swine farm in Jinshan, Shanghai, China. To minimize the impact of particulate matter in the biogas slurry on experimental errors, the slurry was filtered using medium-speed filter paper to remove most particulate solids. The processed slurry was then stored at 4 °C for subsequent use. The characteristics of the row biogas slurry (RBS) are shown in Table S1.

2.2. Experimental design

Biogas slurry pretreatment experiments (Fig. 1) were conducted using a 500 mL reactor in which the RBS was adjusted to pH = 5 and treated under different conditions. Two systems were spiked with 50 g/L of Fe-C particles, and the air was passed (50 mL/min) in the form of microbubbles and conventional bubbles, which were recorded as MFC and CFC, respectively. Another system without Fe-C particles was fed only with air microbubbles (50 mL/min), recorded as MBs. All systems were maintained at room temperature (25 °C) for 2 h, and samples were taken at regular intervals during the reaction for analysis.

The treated biogas slurry was used to culture microalgae to explore improvements in microalgae growth. The microalgae (*Chlorella pyrenoidosa*, FACHB-9) was consistent with previous study [25]. To optimize microalgae cultivation, the processed biogas slurry was appropriately diluted and supplemented with phosphorus after pretreatment. Detailed information is presented in Text S2.

2.3. Analytical methods

2.3.1. Determination of biogas slurry water quality

The analytical methods of water quality are illustrated in the Text S3. 3D excitation − emission matrix (EEM) spectrofluorometry (Text S4) and two-dimensional correlation spectroscopy of UV (UV 2D-COS) (Text S5) were used to characterize the fluorescent and UV-active components, respectively. Several optical parameters were calculated from the EEM and UV − visible spectral data are outlined in Table S2. A principal component analysis (PCA) was conducted using the Origin 2021.

2.3.2. Characterization of Fe-C micro-electrolysis rate

The Fe-C particles after the reaction were collected and then washed with deionized water. After vacuum freeze-drying, surface samples of

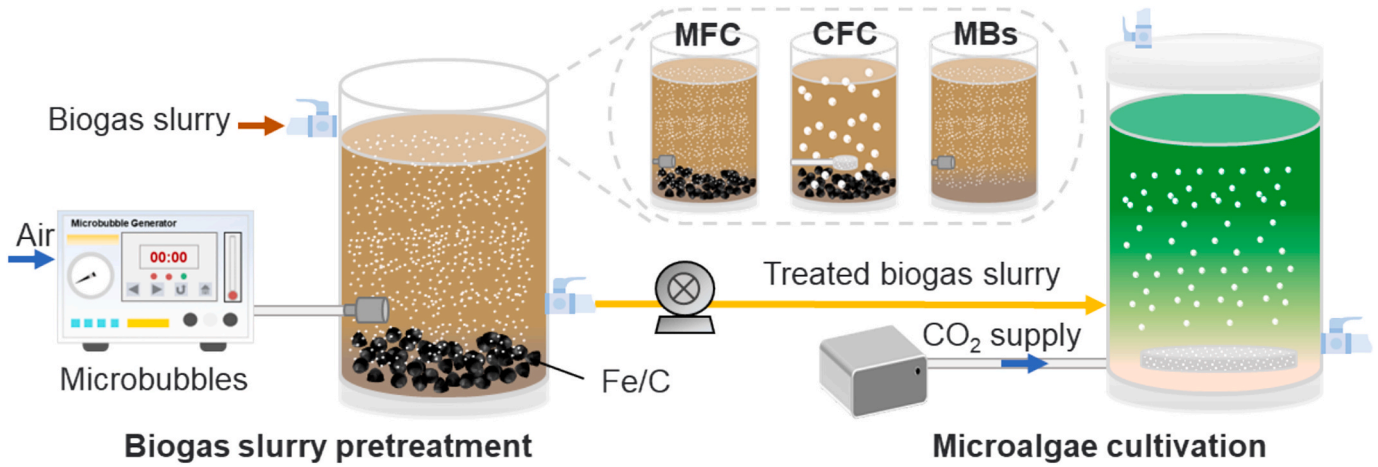


Fig. 1. Schematic showing assembly of the experimental system.

Fe-C particles were scraped and ground with an agate mortar and then characterized (Text S6). The electrochemical properties of the Fe-C particles were measured using a three-electrode cell (CHI760E, CH Instruments, China). Details of the electrode preparation methods and tests are given in Text S7.

2.3.3. Determination of active species

Fe (IV) was detected by the sulfoxide probe method [19]. The amount of •OH is determined by terephthalic acid photoluminescence (TA-PL) probing means. The detailed methodology is in Text S8.

2.3.4. Characterization of floc samples

Floc samples were centrifuged (3000 rpm, 2 min) and freeze-dried to obtain a powdered solid. The surface functional groups and elemental composition analysis of floc samples can be found in Text S9. Synchrotron radiation-based Fourier transform infrared (SR-FTIR) obtained at the beamline BL01B1 of the National Facility for Protein Science in Shanghai (NFPS) was used to analyze differences in the distribution of different functional groups, the details are given in Text S10. Extended x-ray absorption fine structure (EXAFS) of the coordination environment of Fe within flocs was analyzed by bulk Fe K-edge (7112 eV) on beamline BL14W1 of the NFPS, the details are shown in Text S11.

2.3.5. Determination of microalgae growth and cell activity

The microalgae biomass concentrations were determined by optical density measurements at a wavelength of 680 nm [25]. Microalgae cultured for 3 days were taken to measure the intracellular oxidative stress activity [35]. Details of the methodology are in Text S12. The fluorescence of the microalgae was measured using a chlorophyll

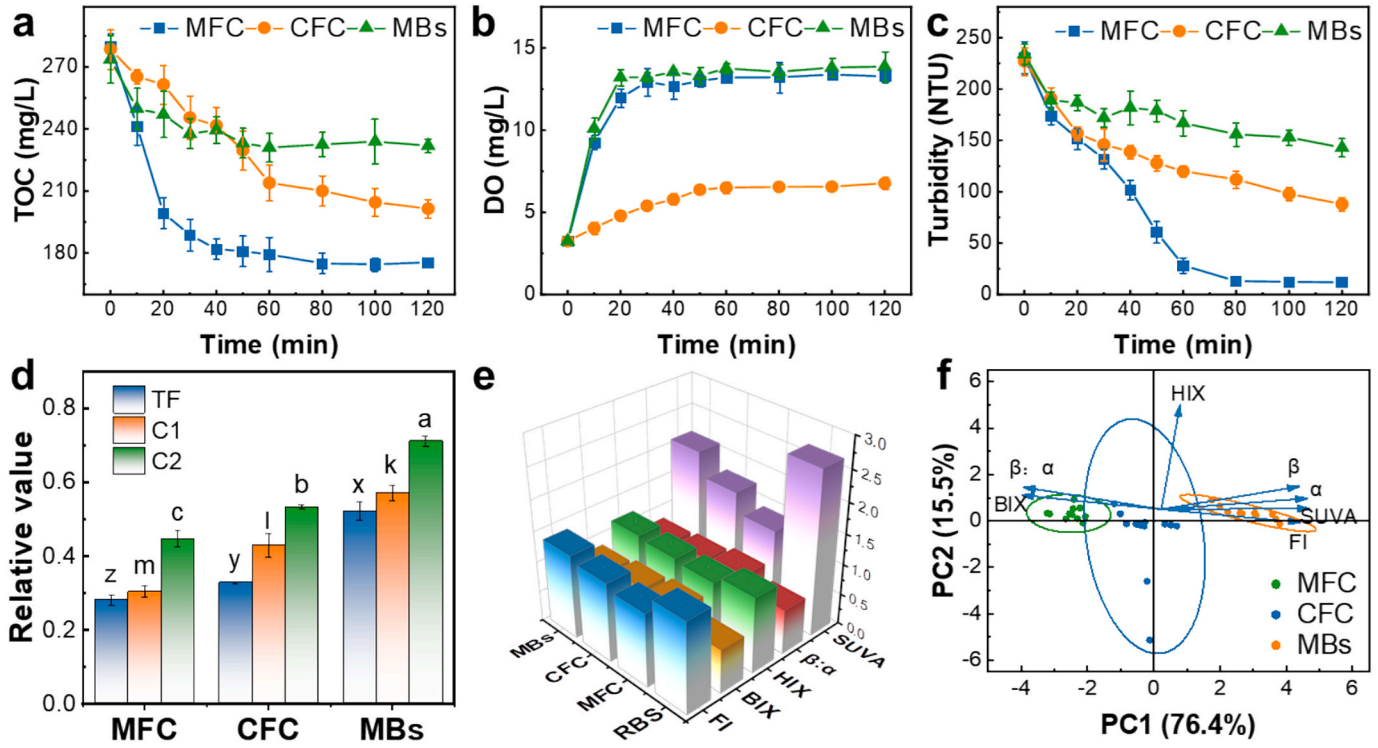


Fig. 2. Variation in (a) TOC, (b) DO, and (c) turbidity of the biogas slurry with operation time. (d) The relative value of TF and two fluorescence components (C1 and C2) of different systems. (e) Variation in SUVA, β: α, HIX, BIX, and FI of different systems. (f) PCA results show the difference among MFC, CFC, and MBs systems, data points are samples at 60, 80, 100, and 120 min. (The same words indicate no significant difference, $p > 0.05$).

fluorescence meter (AquaPen AP110-C, Czech Republic).

2.4. Statistical analysis

All data were evaluated with IBM SPSS statistics 25.0 software using ANOVA to compare the data of each system for significant change (95 % confidence interval). Significant difference was identified at the levels of $p < 0.05$.

3. Results and discussion

3.1. Effects of degradation and coagulation on water quality

The optimization of microbubbles-assisted pretreatment performance was evaluated by varying water quality characteristics in different systems. The total organic carbon (TOC) concentration (Fig. 2a) in the MFC system decreased drastically within the initial 20 min and continued to decrease moderately from the 20th to the 120th minute. The MFC system presented the highest TOC removal efficiency (37.3 %), followed by the CFC system (27.7 %) and the MBs system (15.3 %). The variation in UV254 (Fig. S3a) was similar to TOC concentration changes across all systems. These findings imply that the DOM elimination ability was notably promoted with the assistance of microbubbles. Microbubbles can generate a larger Laplace pressure and exhibit extended retention time in the water, allowing them to increase the O_2 transfer rate [8]. Consequently, we observed a significant increase in dissolved oxygen (DO) concentration (Fig. 2b) in the MFC and MBs systems. As an electron acceptor, O_2 can increase the cathodic reduction reaction potential, and its increased concentration can accelerate electron transfer and DOM degradation [39]. Additionally, the production of •OH by the collapse of microbubbles [9]may contribute to the degradation of DOM [8].

The Fe (hydr)oxides generated by the iron electrode reaction can further remove DOM and suspended particulate matter by adsorption and coagulation [33]. This can be verified by the substantial flocs formed in the MFC and CFC systems (Fig. S4). Compared with the CFC system (61.2 %), the turbidity (Fig. 2c) removal efficiencies were improved by 55.0 % in the MFC system (94.9 %), indicating that coagulation and settling properties of particulate matter with Fe (hydr) oxides were facilitated by the microbubbles. The variation in TN, NH_4^+-N, and PO_4^{3-}-P concentration is illustrated in Fig. S3. PO_4^{3-}-P was rendered undetectable in the MFC and CFC systems after the operation due to the formation of $Fe_3(PO4)_2$ and $FePO_4$ precipitates [33]. Detailed discussion about the effects of different systems on TN, NH_4^+-N, and PO_4^{3-}-P removal is available in Text S13.

3.2. Specific types of DOM elimination analysis

The components of DOM in the biogas slurry are complex, and different components exert distinct effects on biochemical and microalgae activities. Different DOM removal efficiencies in various systems were further explored. The fluorescence properties of DOM changed substantially after treatment (Fig. S5). The total fluorescence (TF) of DOM decreased to 28.2 %, 32.9 %, and 52.5 % of RBS in the MFC, CFC, and MBs systems, respectively (Fig. 2d). The parallel factor (PARAFAC) analysis identified two independent fluorescence components: humic-like (C1) and protein-like (C2) substances (Fig. S6 and Table S6). The MFC system exhibited escalated removal of both components (Fig. 2d). Notably, C1 demonstrated a higher removal rate compared to C2, suggesting it is more easily dominated in electron competition [18]. Fluorescence regional integration (FRI) results also showed that the C1 component (Region V) in the MFC and CFC systems is lower than that in the MBs system (Fig. S8). Previous studies have reported that C1 can be removed by physical processes such as adsorption [33,37]. Thus, electron transfer and highly reactive Fe mineral production in the system can be enhanced by microbubbles, favorably contributing to C1 removal. The removal of C2 in the MFC system could be attributed to the improved charge effects.

The relevant optical parameters of biogas slurry were further used to assess the contribution of different types of DOM. The specific UV absorbance (SUVA) describes the aromatic content in humic substances, typically associated with electron-rich and light-absorbing moieties containing C = C and C = O double bonds [18]. The results (Fig. 2e) showed that the SUVA of the MFC system is significantly lower than the other systems ($p < 0.05$). The heightened generation of iron ions by the MFC system facilitated the elimination of aromatic substances as the C = C and C = O can increase the sites for complex metal ion binding [16,33]. This phenomenon further contributed to the biochemistry increase of biogas slurry [6]. The fluorescence index (FI) is used to identify the relative contribution of land-sourced and microbial-sourced DOM in C1. The MFC system exhibited a lower FI value (<1.4) than the CFC system, indicating that the degradation of microbial-sourced metabolites contributes more to DOM removal. The biological index (BIX) indicates the level of autotrophic productivity, implying heightened bioavailability with elevated values [18]. MFC and CFC systems showed higher BIX than MBs systems due to the higher removal ratio of C1. The freshness index (β:α) indicated that only the MFC system significantly differed from RBS ($p < 0.05$), suggesting more soluble compounds were produced during degradation.

As for the UV-active substances, UV 2D-COS was used to derive removal mechanisms in different systems (Fig. S10). The UV-synchronous spectra illustrate that more UV-active substances are removed in the presence of microbubbles [37]. UV-asynchronous spectra suggest that the reduction of benzenoid was more sensitive with microbubbles. Detailed discussion is available in the Text S14. PCA results showed that the explained variance of the samples separated in the first and second principal axes for each system was 76.4 % and 15.5 %, respectively (Fig. 2f). Indicators representing DOM content (i.e., SUVA and TF) were clustered at the positive side of the first principal axis. Their narrow-angle relative to the first principal axis contributed to better sample separation. The sample points of the MFC system were distributed on the negative side of the first principal axis, indicating that MFC systems exhibited superior DOM removal than CFC and MBs systems. The indicators representing the autochthonous DOM (i.e., BIX and β: α) were clustered at the negative side of the first principal axis. Their positions in the load plot corresponded closely to the positions of the samples in the MFC system of the score plot, suggesting that the MFC system produces more soluble small molecule DOM.

3.3. High oxidizing properties accelerate electron transfer

The degradation efficiency of DOM was correlated with the electron transfer rate of Fe-C micro-electrolysis [15], so the effect of microbubbles on the electron transfer rate in the MFC system was further investigated. The electron transfer rate can be reflected by the corrosion of Fe. Analysis via SEM revealed the corrosion of Fe on the surface of Fe-C particles. Compared to the raw Fe-C particles, the surface of Fe-C particles from both the MFC and CFC systems (Fig. 3) exhibited increased roughness and porosity, where the corrosion of Fe causes the creation of voids on the surface. Notably, the surface of Fe-C particles from MFC systems was covered by more tiny floc particles, indicating more pronounced corrosion of the Fe [39]. This phenomenon was attributed to microbubbles with extensive diffusion capability, penetrating the micropores of the Fe-C particles and intensifying the oxidative dissolution of Fe. Previous studies have demonstrated that the dissolved Fe can form nanoparticles, promoting the formation of tiny floc [33]. Elemental distribution (Fig. 3) shows a higher quantity of Fe on the surface of Fe-C particles in the MFC system (14.91 %) compared to the CFC system (9.16 %), indicating that more dissolved iron ions are reabsorbed to the surface.

The electron transfer process at the anode consumes H^+ causing an increase in pH. The pH value measured approximately 7.13 in the MFC

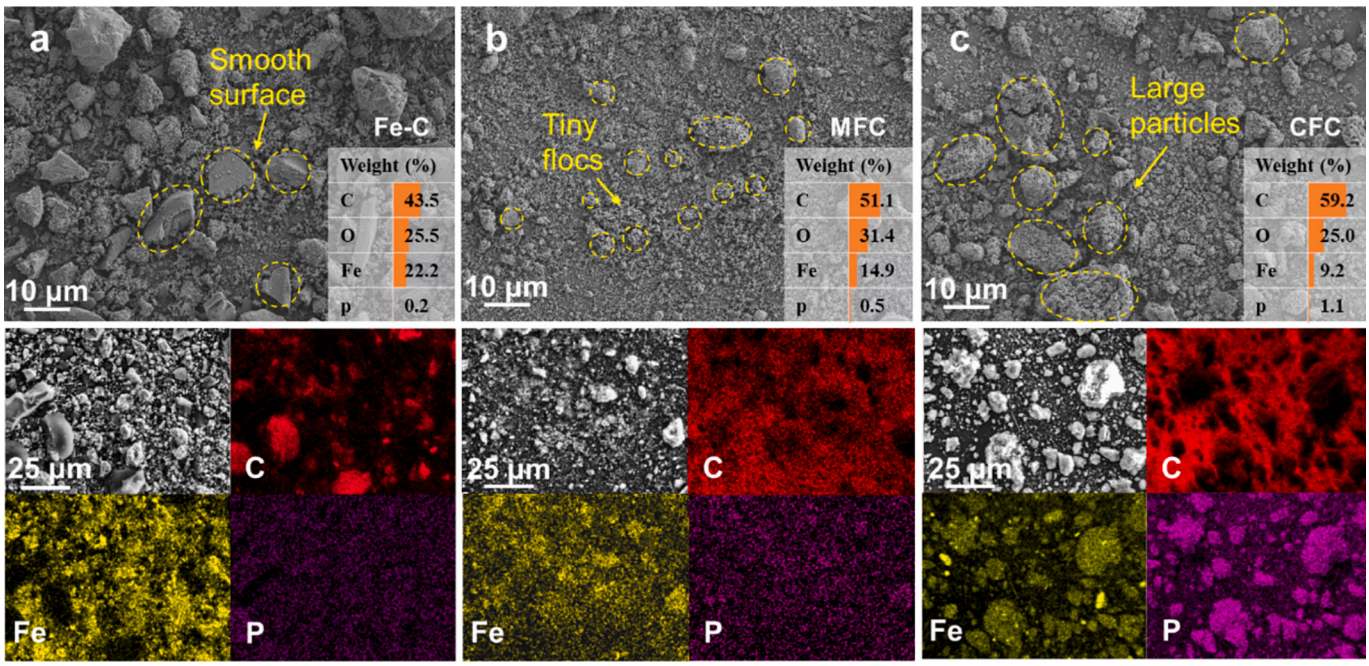


Fig. 3. SEM and distribution of the elements C, Fe, and P of different Fe-C particles. The values in the table are the surface elements weight ratio analyzed by EDS.

system after operation (Fig. 4a). The MFC system was more alkaline compared to the CFC system, confirming that the microbubbles accelerated the electron transfer rate in the system. Additionally, XRD demonstrated that peaks of Fe and FeO in Fe-C particles of the MFC system were lower than the CFC system (Fig. S12a). These results suggest that microbubbles enhanced the rate of micro-electrolysis, resulting in increased dissolution of iron ions. However, we found that iron ions were significantly lower ($p < 0.05$) in the MFC system than in the CFC system after operation (Fig. 4c). This suggests that the high oxidation rate of microbubbles accelerates the oxidation of Fe(II) to Fe(III) and the subsequent formation of insoluble Fe minerals. Dissolved iron ions in the CFC system primarily exist as Fe(II) due to the acidic environment reducing the oxidation rate of Fe(II) [39]. This also theoretically explains the superior adsorption and coagulation capacities for pollutants

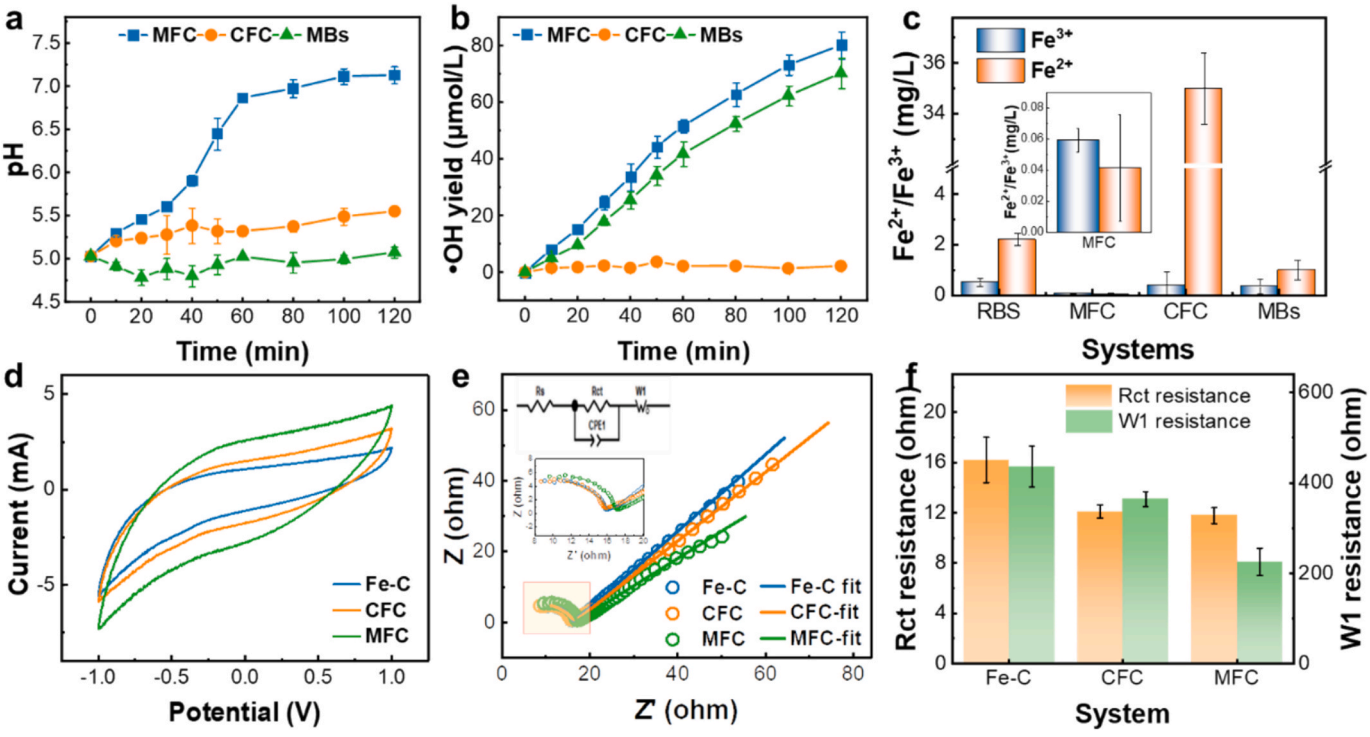


Fig. 4. (a) Variation in pH of the biogas slurry. (b) Detection of •OH yield in different systems. (c) The concentration of iron ions in the supernatant of RBS and different systems. (d) Cyclic voltammograms and (e) Nyquist plots of EIS for raw Fe-C particles and Fe-C particles in MFC and CFC system (The same equivalent circuit model was used for the three systems in fitting the EIS curves. It consists of electrolyte resistance (Rs), charge transfer resistance (Rct), constant phase element (CPE), and Warburg impedance (W1).). (f) Resistance for the fits before and after the reaction of the Fe-C particles.

in the MFC system compared to the CFC system.

The generation of active species has a non-negligible contribution to electron transfer and DOM degradation. Microbubbles can generate •OH to induce the oxidative transformation of Fe and DOM [8]. Fig. 4b illustrates minimal •OH formation in the CFC system. In contrast, •OH production increased over time in the MFC and MBs systems, indicating that the microbubbles can effectively promote •OH generation. The higher levels of •OH in the MFC system compared to the MBs system could be attributed to the presence of persistent free radicals (PFRs) in the carbon [27], which promotes •OH production [4]. Previous studies have also reported that Fe minerals are capable of reducing oxygen to produce •OH [21]. Additionally, Fe(IV) can degrade DOM. As shown in Fig. S12b, there is almost no formation of Fe(IV) in any system, indicating the degradation of DOM by Fe(IV) can be neglected.

To reveal the impact of corrosion products and adsorbed particulate matter on electron transfer during Fe-C micro-electrolysis, the electrochemical properties of Fe-C particles were analyzed. The cyclic voltammogram curve (Fig. 4d) illustrates that the double electric layer capacitance of the Fe-C particles in the MFC system is larger than that of the CFC system, implying that the microbubbles may facilitate the formation of additional electrochemical active sites. The Nyquist plots of electrochemical impedance spectroscopy (EIS) are shown in Fig. 4e. Based on the fitting results (Fig. 4f), the charge transfer resistance (Rct) of Fe-C particles in the MFC system (11.8 O) is lower compared to both the raw Fe-C particles (16.2 O) and CFC system (14.1 O), which is consistent with previous studies [18]. Rapid micro-electrolytic reactions and corrosion occurred in the MFC system with the assistance of microbubbles, and the corrosion products formed on the surface accelerated the interfacial charge transfer between the solution and the Fe-C particles. The Warburg impedance (W1) of the Fe-C particles in the MFC system (225.4 O) is also lower than that of both raw Fe-C particles (436.4 O) and the CFC system (363.9 O). This suggests a reduced diffusion resistance of ions in the solution within the voids of the Fe-C particles in the MFC system, enhancing the reaction rate. This could be attributed to microbubbles potentially fully oxidizing the iron within these voids and subsequently enhancing ion diffusion.

3.4. DOM removal by formation of stable structures in flocs

Flocs formation plays a key role in DOM removal. Due to rapid electron transfer leading to the formation of more Fe (hydr)oxides, a higher quantity of flocs (Fig. 5a) was formed in the MFC system (2907.4 mg/L) compared to the CFC system (1387.1 mg/L). Nearly all generated Fe in the MFC system was bound within the flocs, whereas 35.0 mg/L of iron ions remained in the supernatant of the CFC system (Fig. 5b). Faster electron transfer and settling rates result in flocs with higher Fe content in MFC systems compared to CFC systems (Fig. 5c). This implies that the formation of additional Fe (hydr)oxides and stable DOM binding sites in the MFC system facilitated the simultaneous removal of Fe and DOM.

To reveal the binding modes of DOM in flocs, the organic functional groups within the flocs were analyzed. Fourier Transform Infrared Spectroscopy (FTIR) results (Fig. 5e) showed the binding modes of DOM in flocs. The peak at 3412 cm^{-1} arises from the − OH vibration, representing the contribution of absorbed water and organic hydroxyl groups [33]. 2927 cm^{-1} corresponds to the asymmetric (νC-H asym) stretch of the aliphatic compound [4]. Vibration at 1644 cm^{-1} arises from the carbonyl group (C = O) of aromatic rings in the aromatic compound [43], while 1545 and 1385 cm^{-1} were assigned to asymmetric (νC-O asym) and symmetric (νC-O sym) stretching motions of the adsorbed carboxyl group, respectively [34]. The band at 1038 and 470 cm^{-1} was identified as the characteristic vibration of C-O and Fe-O lattice stretching modes, respectively [41]. Some of the peaks of the flocs in the MFC system showed changes compared to the CFC system, indicating possible differences in the type and abundance of DOM bound by the flocs. Furthermore, SR-FTIR analysis was conducted to assess the distribution of organic functional groups within the flocs, confirming substantial differences between the two systems (Fig. 5f). Specifically, the alterations in C = O (1644 cm^{-1}), νC-O sym (1385 cm^{-1}), and C-O (1038 cm^{-1}) indicated increased interactions of carboxyl and carbonyl groups with edges of flocs [34], probably because microbubbles increase the positive surface charge of the Fe (hydr)oxides. Furthermore, the band change at 1038 cm^{-1} was the most pronounced and shifted to the higher frequency, indicating that the flocs were mainly combined with DOM through inner-sphere adsorption, possibly the carbonyl group in the DOM forms a covalent bond with the Fe (hydr)oxides [33].

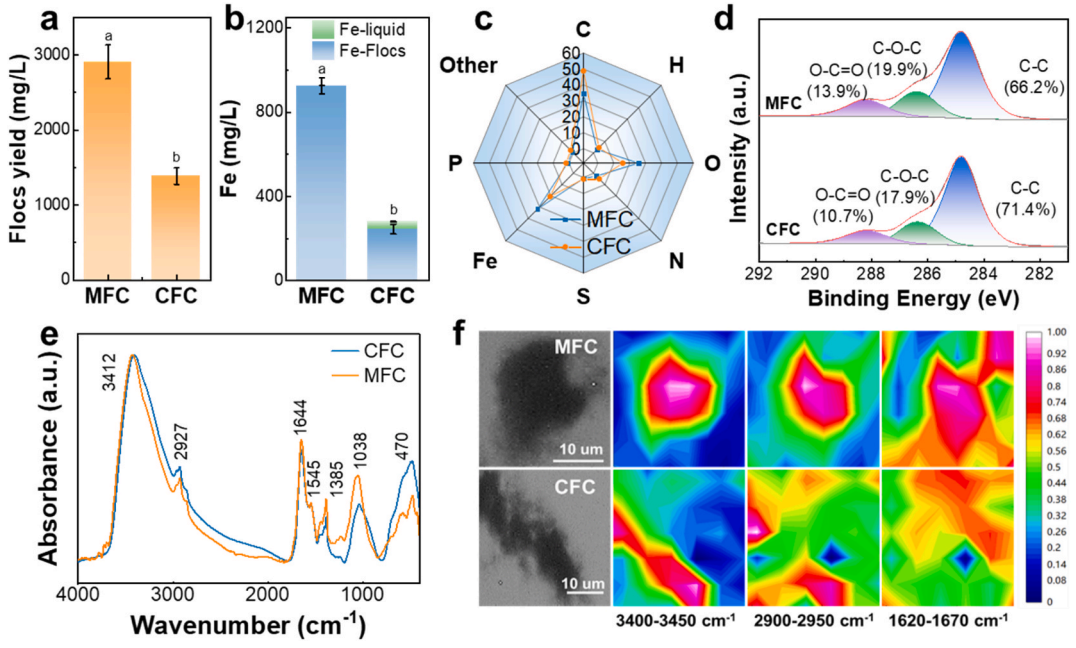


Fig. 5. (a) Flocs yield of MFC and CFC system after operations. (b) Distribution of iron in floc and supernatant after operation. (c) Percentage of different elements in flocs of MFC and CFC after operations. (d) The background-subtracted high-resolution XPS spectra of C1s in flocs. Binding modes of DOM on flocs through (e) FTIR and (f) SR-FTIR analysis (normalized).

The flocs on the composition and proportion of elements based on X-ray photoelectron spectroscopy (XPS) further verified the positive effects of microbubbles on DOM binding modes. Peaks corresponding to Fe2p, O1s, C1s, and P2p were observed in flocs (Fig. S13a). The high-resolution C1s spectra (Fig. 5d) showed peaks concentrated around 284.8, 286.3, and 288.2 eV, representing C–C/C–H bonds in hydrocarbon compounds, C–O–C/C–O bonds in carbonyl groups, and O–C=O/O–C–O bonds in carboxyl-containing substances, respectively [33]. The primary forms of organic matter in flocs were aromatic carbon, α-carbon, and carboxyl carbon. The characteristic peaks of both C–O–C/C–O and O–C=O/O–C–O bonds exhibited a shift toward higher binding energy in the MFC system compared to the CFC system. This phenomenon results from the interaction between Fe minerals and organic matter, altering the electrostatic potential field of the core electrons of the C atom [16]. The proportion of α-carbon and carboxyl carbon in the MFC system exceeded that in the CFC system. This suggests that more organic matters were bound to flocs through carboxyl and carbonyl groups via electrostatic interactions, aligning with FTIR results. Studies have shown that the carboxyl group on DOM is partially ligand-complex with the –OH groups on Fe (hydr)oxides to form inner-sphere complexation [33]. The Fe2p spectra of samples are shown in Fig. S13b. The higher Fe (III) content in the CFC system than in the MFC system suggests that the microbubbles possessed a greater oxidizing capacity, enhancing the binding capability of flocs with organic matter. Furthermore, the Fe/C ratio (Fig. 5c) implies that the ability of unit Fe atoms to bind C atoms is higher in the MFC system (0.92) than CFC system (0.42).

3.5. Accelerated coagulation by the accumulation of ferrihydrite

The amorphous Fe mineral (e.g., ferrihydrite) in flocs hinders further research by conventional means. To further understand the effect of floc structure on coagulation, the mineral structure and coordination environment of Fe within flocs were examined using EXAFS. Linear combination fitting (LCF) showed that the flocs were predominantly ferrihydrite, accounting for 84.8 % and 66.3 % of the MFC and CFC system, respectively, the rest were lepidocrocite and maghemite. This indicates that the microbubbles impede the transformation of ferrihydrite to highly crystalline minerals. The Wavelet transform (WT) analyses of reference minerals and floc structures are shown in Fig. 6 and Fig. S14. The signals at R = 1.46 Å and R = 2.57 Å were observed in two samples, indicating coordination bonds in the first and second shell layers, respectively. The peak positions of the coordination bonds were close to the reference minerals, meaning that the coordination bonds in the first and second shell layers are dominated by Fe-O and Fe-Fe paths,

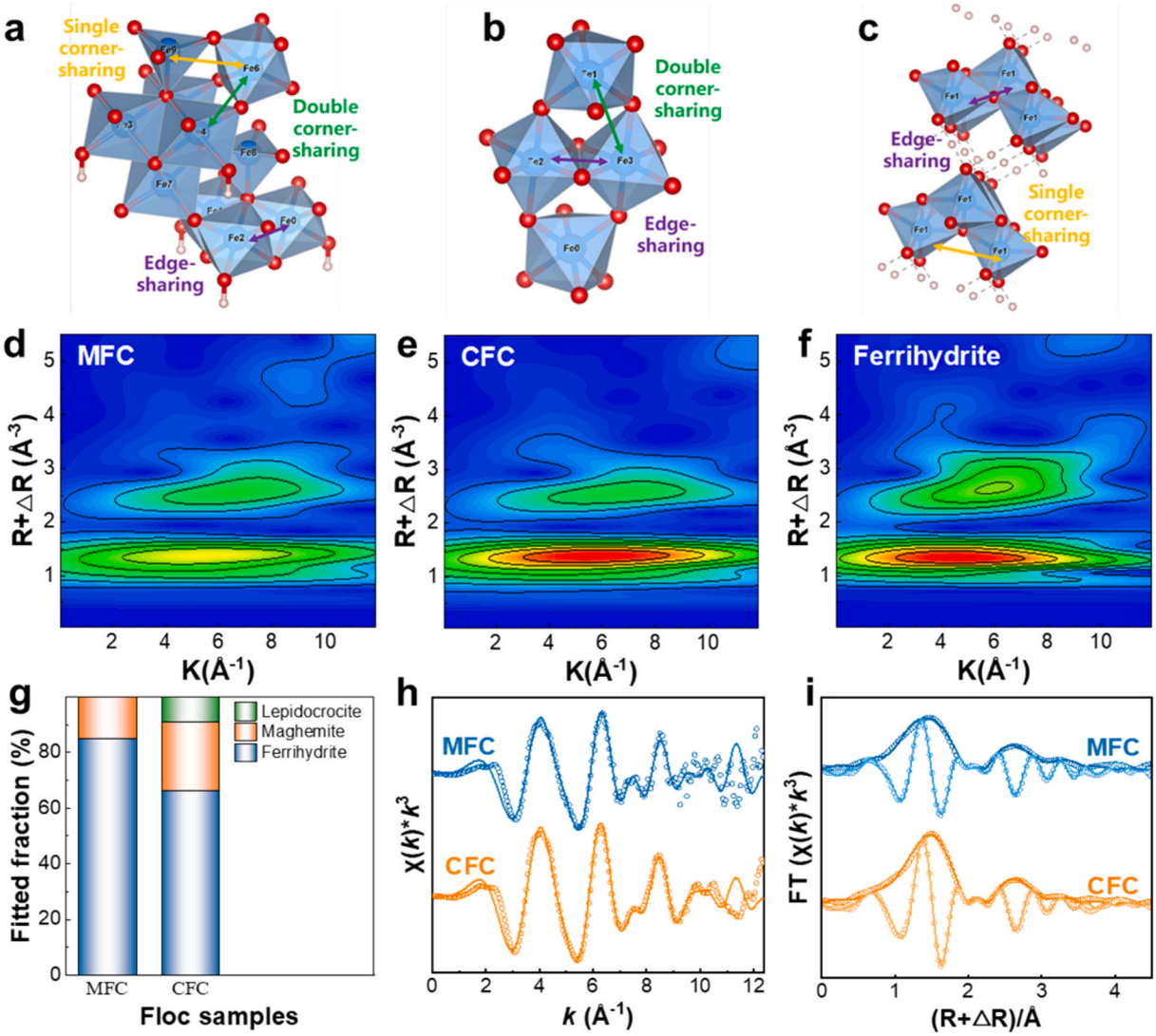


Fig. 6. Crystal models of reference minerals: (a) ferrihydrite, (b) maghemite, and (c) lepidocrocite. Wavelet transform plots of flocs formed by (d) MFC system, (e) CFC system, and (f) standard mineral of lepidocrocite. (g) LCF results of EXAFS. (h) Fe k3 −weighted EXAFS spectra and (i) their Fourier-transform imaginary parts and magnitudes of floc samples.

respectively. The relative peak intensity at R+△R = 1.46 Å showed that flocs formed in the MFC system had a lower intensity than the CFC system. This indicated that Fe in the MFC system exhibited a reduced affinity for combining with O atoms, resulting in a lower proportion of Fe-O coordination.

The local coordination environment of Fe in flocs was explored by shell-fit analyses of k3-weighted Fe K-edge EXAFS spectra. Two Fe-O paths were required to accurately model the first O shell with the same Debye-Waller parameters (σ^2). The Fe-O1 (1.93 Å) and Fe-O2 (2.00 Å) paths represent FeO4 tetrahedra and FeO6 octahedra, respectively [22]. In subsequent fits extended to 4.0 Å (R+△R), the σ^2 of Fe-O paths were fixed to reduce the variables. The final fit model contained three additional Fe-Fe paths representative of edge- (Fe-Fe1), double corner- (Fe-Fe2), and single corner-sharing Fe (Fe-Fe3) [28].

The final model fits are illustrated in Fig. 6 and the corresponding EXAFS parameters are summarized in Table S7. For the first shell, the coordination numbers of Fe-O1 and Fe-O2 are 2.0 and 2.4 in the MFC system and 2.3 and 2.9 in the CFC system, respectively. This result suggests that more proportional FeO4 tetrahedra were formed in the MFC system (45.5 %), leading to a decrease degeneracy (N) sum of the Fe-O paths. Additionally, the reduced Fe-O distance distributions lead to significant octahedral distortions, resulting in phase-canceling interference of the individual Fe-O scattered waves [28]. For the iron shells, the flocs in the CFC and MFC systems have the same number of Fe1 atoms. The number of Fe2 atoms in the flocs of the MFC system is lower than that of the CFC system. Surface negatively charged microbubbles can interact with Fe(III) [8]. The continuous perturbation of microbubbles might disrupt the crystal structure and inhibit the double corner coordination between FeO6 octahedrons. The disorder of FeO6 octahedrons might inhibit further transformation of ferrihydrite. The accumulation of ferrihydrite could readily adsorb DOM in the biogas slurry, due to its remarkable sorption capacity and high surface area [5]. With the introduction of microbubbles, the oxidation rate of Fe was accelerated, resulting in faster precipitation. Consequently, the contribution of single corner-sharing Fe increased rapidly, which favors the formation of more single corner-sharing dimers. Although lepidocrocite has a single-corner

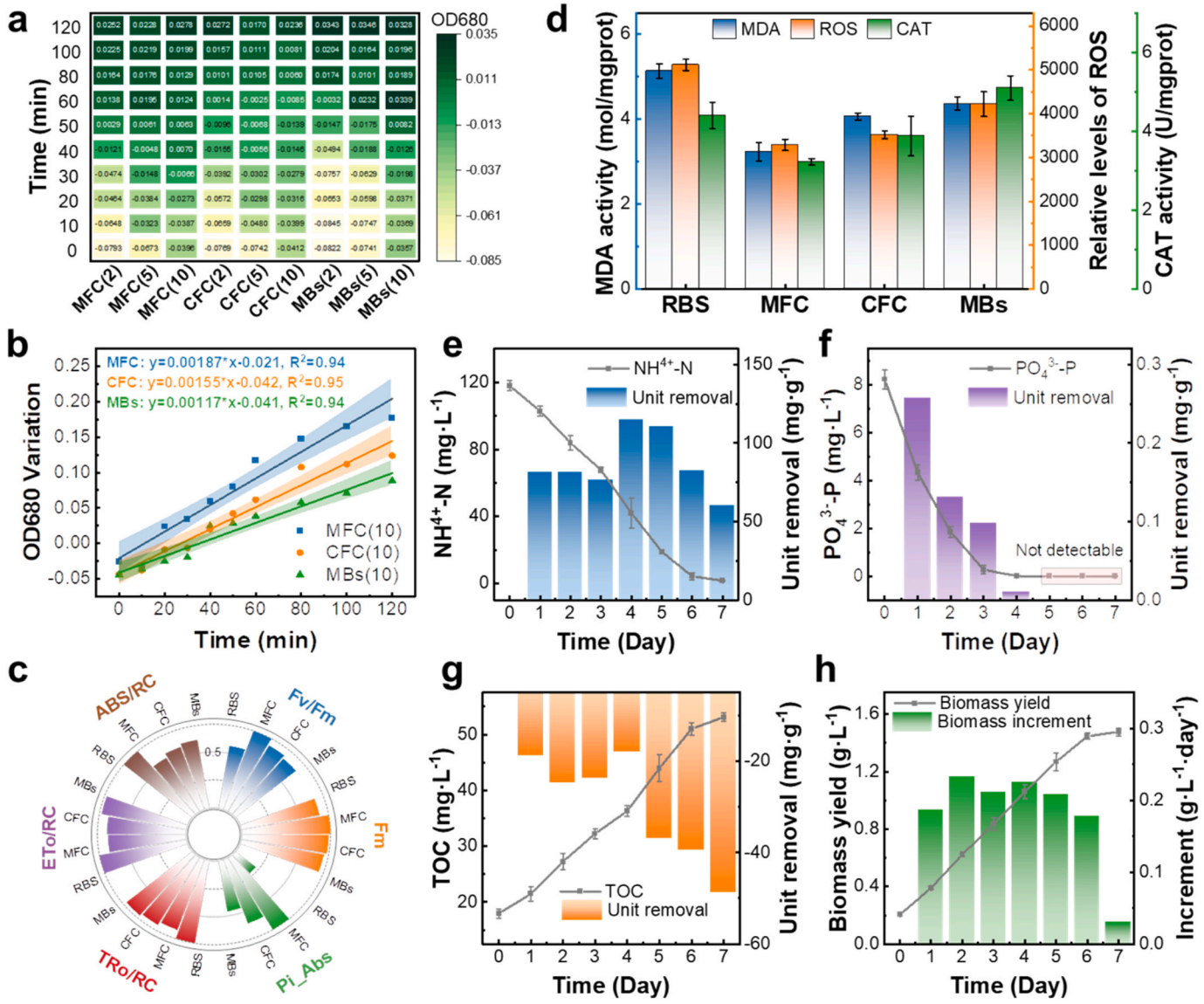


Fig. 7. (a) Microalgae cultivation by biogas slurry at different operation times. Values in parentheses are dilutions (2×, 5×, and 10 ×) of the treated biogas slurry. (b) After phosphate supplementation, microalgae growth was fitted in different systems of 10-fold dilution of biogas slurry. (c) Changes in chlorophyll fluorescence-related index of microalgae cultured in different systems (Indicators were normalized to facilitate comparisons). (d) Changes in MDA, ROS, and CAT in microalgae grown in the biogas slurry from different treatment systems. Microalgae for the utilization of pollutants in biogas slurry after pretreatment (10-fold dilution, supplemented with phosphate: 8.4 mg/L): Removal of (e) NH_4^+-N, (f) phosphate, and (g) TOC by microalgae. (h) Changes in biomass and daily increments of microalgae.

sharing (Fe-Fe3) structure with a high coordination number, there is no lepidocrocite present in the flocs of the MFC system. This suggests that the single corner-sharing dimer is mainly contributed by ferrihydrite. This is due to the lamellar structure of lepidocrocite, where continuous perturbation and oxidation caused by microbubbles are not conducive to the transformation of ferrihydrite to the highly crystalline lepidocrocite. Therefore, the rapid oxidation of microbubbles leads to changes in FeO6 octahedral polymerization morphology, particle size, and morphology. More dispersed disordered nanoscale polymers were formed due to the incorporation of more single-corner sharing (Fe-Fe3) structures. These polymers can promote the nucleation growth of ferrihydrite and enable self-accelerating coagulation.

3.6. Specific pollutant removal improves microalgae cultivation

The treated biogas slurry of all three systems promoted the growth of microalgae compared to RBS (Fig. 7a). To regulate the nutrient ratios, 8.4 mg/L phosphate (TAP medium phosphate solution) was further supplemented to the biogas slurry (NH_4^+-N/PO_4^{3-}-P = 14:1 after supplementation). The biomass of the MFC system increased by 42.7 % and 101.1 % compared to the CFC and MBs systems, respectively, after 10-fold dilution of the biogas slurry (Fig. 7b). This suggests that the MFC system greatly improves the microalgae growth environment by reducing turbidity and removing microalgal stressors (e.g., humic-like substances, aromatic compounds, and residual iron ions) from the biogas slurry.

The removal of microalgal stressors from biogas slurry in different systems was reflected by the oxidative stress reactions in microalgae cells (Fig. 7d). The increase in these biomarkers implies that microalgae cells may be subjected to oxidative stress damage. Previous studies showed that high concentrations of humic acid can penetrate microalgae cells, damage cell membranes, and thereby cause oxidative stress [40]. After pretreatment, the oxidative damage of microalgae cells was alleviated, especially in the MFC system with the lowest reactive oxygen species (ROS), malondialdehyde (MDA), and catalase (CAT) levels. This indicates that the reduction of harmful substances such as humic acid in biogas slurry after pretreatment can effectively improve the growth of microalgae. Furthermore, hydrophilic functional groups (e.g., carboxyl groups) in aromatic compounds can bind with hydroxyl groups in extracellular polymeric substances (EPS) via H-bonds [3], which can cause oxidative stress. Section 3.3 has demonstrated that the carboxyl group in the biogas slurry is involved in the binding of flocs and Fe minerals and can be effectively removed. Furthermore, ROS can also be generated due to abiotic stress from exposure to metal ions. In the CFC system, the concentration of remaining iron ions in the supernatant reached 35.0 mg/L, surpassing the threshold conducive to microalgae growth, potentially inducing oxidative stress to a certain extent.

Additionally, oxidative damage might adversely affect microalgae photosynthesis. The following biophysical parameters (Fig. 7c) were determined: the maximum quantum yield (Fm) and conversion efficiency (Fv/Fm) of primary photochemistry; specific energy flux per reaction center (RC), including absorption energy (ABS/RC), electron transport energy (ETo/RC), trapping energy (TRo/RC); performance index (Pi_Abs) of light energy absorption. The values of Fv/Fm, Fm, and Pi_Abs generally decreased with increasing levels of oxidative stress, whereas the values of TRo/RC, ETo/RC, and ABS/RC were positively correlated with oxidative stress levels. In addition to the reduced oxidative stress, the MFC system greatly reduces the turbidity of the biogas slurry and increases the light transmittance, thereby significantly improving photosynthetic performance. The results indicate that the pretreatment of the biogas slurry increased the light conversion efficiency in the photochemical reaction center and the energy absorption parameter. Compared with the RBS (0.455), the Fv/Fm values in the MFC (0.556), CFC (0.521), and MBs (0.507) systems were increased by 22.2 %, 14.5 %, and 11.4 %, respectively, while the Pi_Abs values were increased by 111.1 %, 81.5 %, and 50.0 %, respectively. Microalgal stressors in the biogas slurry interfered with enzyme activity and photosynthesis on chloroplasts, leading to the down-regulation of major proteins (PsbB, PsbC, and PsbD). Furthermore, ROS generation could disrupt the chloroplast structure and membrane lipid peroxidation homeostasis, ultimately restricting photosynthetic electron transport [12]. Consequently, the absorption (ABS/RC), transfer (ETo/RC), and dissipate (DI0/RC) energy per RC increased due to the inactivation of some RCs [17].

The nutrient utilization in the biogas slurry from the MFC system by microalgae was further investigated. The NH_4^+-N concentration decreased to 1.61 mg/L on day 7 with a removal rate of 98.58 % (Fig. 7e). Phosphate concentration in the supernatant was below the detectable limit level after day 5 (Fig. 7f). The decrease in phosphate concentration did not significantly affect the growth rate of microalgae. Microalgae can absorb excess phosphate, storing it within cells for growth metabolism and energy synthesis in phosphate-deficient external environments. Correspondingly, the biomass of microalgae cultured in the biogas slurry increased to 1.47 g/L after 7 days (Fig. 7h). Interestingly, the concentration of TOC increased from an initial 17.96 mg/L to 53.12 mg/L (Fig. 7g), which is contrary to the results in some studies [42]. It might be attributed to microalgae converting fixed inorganic carbon into organic carbon during photosynthesis, subsequently releasing converted organic carbon into the system as extracellular secretions. Therefore, coupling the MFC system with microalgae cultivation can achieve efficient utilization of nitrogen and phosphorus nutrients in the biogas slurry and promote the conversion and storage of CO_2 to organic carbon.

3.7. MFC-driven microalgae-based biogas slurry treatment mechanism

Based on the above proof-of-concept, we propose the mechanism for MFC-driven oxidative synergistic self-accelerating coagulation to improve microalgae-based biogas slurry treatment (Fig. 8). Microbubbles increase oxygen mass transfer and electron transfer rates to optimize the efficiency of Fe-C micro-electrolysis. Meanwhile, the presence of PFRs in Fe-C particles increased the amount of •OH produced during microbubble collapse [9,27], further accelerating the degradation of pollutants and the generation of Fe(III). The acceleration of the micro-electrolysis process results in an alkaline environment in the system, which is conducive to the production of ferrihydrite with high adsorption properties [30]. Surface negatively charged microbubbles can interact with Fe(III) [8]. Therefore, the continuous perturbation of microbubbles inhibits the transformation of hydrotalcite to a highly crystalline mineral, which leads to the accumulation of hydrotalcite and self-accelerated coagulation. Subsequently, the specific types of DOM (e.g., carboxyl groups) in the biogas slurry were more easily removed by forming a more stable structure in the flocs, thereby reducing the coercive effects on microalgae of binding to functional groups in the EPS [3]. In addition, particulate matter was removed by coagulation, improving the light utilization of microalgae. Efficient removal of these stressors reduced oxidative stress in microalgae and increased the maximum quantum yield of the photosynthetic system. Consequently, nutrients such as nitrogen and phosphorus in the biogas slurry are efficiently utilized by microalgae, and microalgae biomass is notably increased.

4. Conclusions

This study presented a novel MFC system for biogas slurry pretreatment, improving its suitability for microalgae cultivation, as well as breaking through the roadblocks of conventional wastewater resource utilization. The MFC system not only overcame the limitation of electron transfer efficiency in the conventional Fe-C treatment system but also shifted the focus of Fe-C micro-electrolysis research to the coagulation-driven removal of pollutants. The results demonstrated that microbubbles achieved efficient removal of DOM by accelerating the

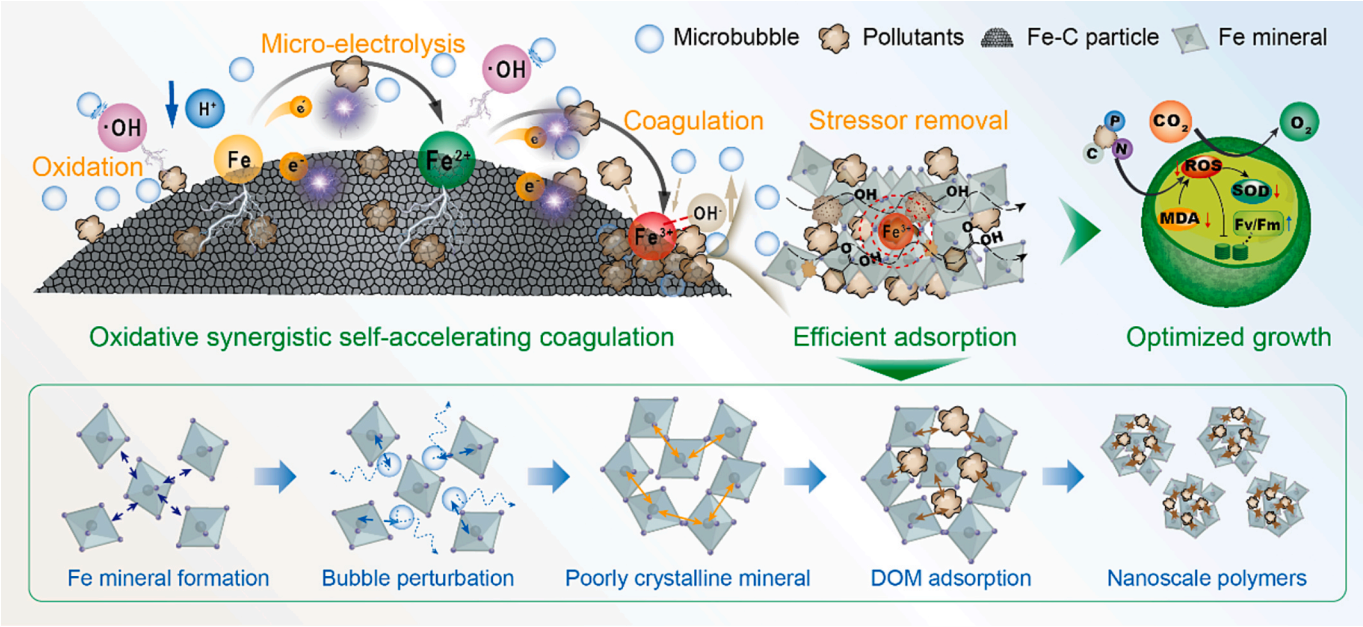


Fig. 8. Proposed mechanism of MFC-driven microalgae-based biogas slurry treatment.

oxidation and coagulation processes, revealing the mechanism of self-accelerating coagulation during Fe-C micro-electrolysis. These findings have critical implications for the application of Fe-C treatment systems in water treatment and the Fe cycle in nature. In addition, the response of parameters such as microalgae oxidative stress and photosynthetic activity to the removal of specific types of DOM in complex systems of biogas slurry was investigated. This provides a reference for the selective removal of biotic stressors in future bioremediation technology studies. Compared with the CFC system, the MFC system can transfer most of the iron ions in the solution to the floc due to better flocculation characteristics, reducing the risk of secondary contamination of dissolved iron ions. Although phosphate transfer to the flocs as strengite or vivianite during the Fe-C micro-electrolysis may diminish the available amount in the supernatant for microalgae growth, subsequent pH adjustments could release these compounds for further utilization. Therefore, coupling the MFC system with microalgae cultivation can achieve efficient utilization of nutrients in the biogas slurry and promote CO_2 sequestration. Furthermore, the MFC system has great potential for scaling up due to ease of operation and low retrofit costs. Previous studies have also indicated that in industrial-scale implementations, microbubbles reduce energy consumption through improved efficiency, lowering total costs by 40 % over a decade [8]. Overall, this study not only inspires future creativity and innovation in the Fe cycle and bioremediation technologies of wastewater treatment but also provides a promising process with great potential for supporting the transition to sustainable water treatment.

CRediT authorship contribution statement

Xiaogang You: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Libin Yang:** Writing – review & editing, Supervision, Methodology. **Mingchao Yang:** Resources, Formal analysis. **Jiaying Ma:** Validation, Formal analysis. **Futao Yi:** Methodology, Formal analysis. **Tongcai Liu:** Writing – original draft, Methodology. **Bo-Yu Peng:** Writing – review & editing. **Jiabin Chen:** Writing – review & editing. **Huaqiang Chu:** Writing – review & editing. **Buchun Si:** Writing – review & editing. **Xuefei Zhou:** Writing – review & editing, Supervision, Funding acquisition. **Yalei Zhang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.164185.

Data availability

The data that has been used is confidential.

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